

BANARAS HINDU UNIVERSITY
B.A. (Hons.) Semester II Examination, 2022-23

Paper No. BAS-201: Algebra, Probability and Statistical Methods

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Define probability of an event. State and prove the addition law of probability for n events. [2+7]
- (b) Two cards are drawn from a pack of ordinary playing cards. Find the probability that the second card drawn is a heart. [5]

Question 2. Explain the following terms with appropriate examples:

- (a) Equally likely events [3]
- (b) Exhaustive events [3]
- (c) Mutually exclusive events [3]
- (d) Random experiment [2]
- (e) Independent and dependent events [3]

Question 3.

- (a) State and prove Bayes' theorem. [7]
- (b) In a bolt factory, machines A, B, and C manufacture 25%, 35%, and 40% of the total output respectively. Of their outputs, 5%, 4%, and 2% respectively are defective bolts. A bolt is taken at random and found to be defective. What is the probability that the bolt came from machine A? [5]
- (c) Define discrete and continuous random variables with appropriate examples. [2]

Question 4.

- (a) Find the probability distribution function for the number of heads when 3 balanced coins are tossed. [4]

- (b) Explain the concept of a random variable. What is a cumulative distribution function (c.d.f.) and what are its properties? [10]
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Question 5.

- (a) Define mathematical expectation of a random variable. [2]
- (b) For what value of A , the function defined below will be a probability density function (p.d.f.)?

$$f(x) = \begin{cases} Ax^2(1-x), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Also find $E(X)$, $E(X^2)$, $Var(X)$, and $P\left(\frac{1}{4} < X < \frac{1}{2}\right)$. [12]

Question 6.

- (a) What is a binomial experiment and what are its properties? Derive the mean and variance of a binomial distribution. [10]
- (b) Let X have a binomial distribution with parameters $n = 4$ and $p = \frac{1}{3}$. Find $P(X = 1)$ and $P(X = 3)$. [4]
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Question 7.

- (a) Describe a Poisson distribution. Derive its mean and variance. [8]
- (b) Derive the recurrence relation for the moments of binomial and Poisson distributions. [6]
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Question 8. Write short notes on any **four** of the following:

- (a) Properties of a good estimator
- (b) Null and alternative hypotheses, and simple and composite hypotheses
- (c) Two types of errors in hypothesis testing
- (d) Any one application of Chi-square statistic in significance testing
- (e) Any one application of F -statistic in significance testing

[4 × 3.5]

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